W3WI_110.2 - Distributed Systems

DHBW
Duale Hochschule
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Mannheim

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Version: 5 (24SCA und 24EG/EH)

Core Topics

- Terminology, concepts, architectures, requirements profiles and architecture models for distributed systems
- Design and implementation approaches
- Comparison of different middleware concepts
- Synchronous and asynchronous communication, remote method calls
- Asynchronous communication and messaging systems
- Security aspects in distributed systems

Previous Knowledge

- Who is familiar with Java?
- Who is familiar with Python?
- Who is familiar with JavaScript/Typescript?
- Who knows what a RESTful API is?
- Who has ever logged in to a server via SSH?
- Who has ever done any administration of a Linux server?
- Who is familiar with Unix/Linux/Mac OS command line tools?
- Who has done any development outside of university projects?

Examination - Portfolio

Background

- the module *Developing Distributed Systems* has 55 lecture hours
- the lecture *Distributed Systems* has 22 lecture hours

  ⇒ Distributed Systems will contribute up to **50** points (out of 120) to the final grade for the module. (Please, don't do the math.)

2 Parts

01 Presentations

02 Programming Exercise

Presentations - General Conditions

- Each person should present for 12 Minutes sharp!
- The presentations should deal in particular with the core content of the lecture and *be of a conceptual nature*.

  This means, after briefly presenting the overall purpose of the framework/technology/protocol, the architecture/the details must be presented. I. e., how errors are dealt with, which services are offered, which guarantees/security aspects are implemented, how scalability is achieved, etc.

  No promotional presentations and no extended motivation presentations! Focus on the topic at hand.
- Presentations have to be in English.
- The speakers should not rotate several times during the presentation. I. e. the first speaker presents first, then the second, and so on. This is necessary for the grading.

Presentations - Submissions

Attention!

You have to upload your presentation to Moodle up until 7 o'clock on the day of your presentation.

Attention!

Further Requirements and Checklist (in Ger.)

The checklist hast to be signed and uploaded together with the presentation. (As a second PDF file.)

Presentations - Available Topics

Hint

Students giving presentations belonging to the same block have to coordinate with each other to avoid any overlap. If you need a specific topic to be covered by another student but are not sure whether it will be presented sufficiently, create a backup slide for your presentation that covers this topic as well and mark it as a backup slide. This backup slide will not be counted towards the time limit.

Virtualization and Virtualization Platforms

01 Introduction to Virtualization & Use Cases

- What is virtualization? Historical context and motivation
- Different types/levels of virtualization
- Key use cases: server consolidation, cloud computing, development/testing, isolation
- Benefits and trade-offs

02 Hypervisors - Architecture & Types

- What is a hypervisor?
- Type 1 (bare-metal) vs Type 2 (hosted) - architectural differences
- Full virtualization vs paravirtualization approaches
- Examples and when to use each type

03 Virtual Machines - Implementation & Management

- VM structure and components (virtual hardware, guest OS)
- VM lifecycle: creation, running, pause/resume, snapshots
- VM migration (live migration concepts)
- Resource allocation and isolation

04 Containers & OS-level Virtualization

- Container concept and how it differs from VMs
- Namespaces and cgroups (conceptual)
- Container images and layering
- Use cases and comparison with VMs

05 Memory Virtualization

- The address translation problem (guest virtual → guest physical → host physical)
- Shadow page tables approach
- Hardware-assisted virtualization (EPT/NPT)
- Memory management techniques (overcommitment, ballooning)

06 Network & I/O Virtualization

- Challenges of virtualizing network and I/O devices
- Emulated vs paravirtualized devices
- SR-IOV (Single Root I/O Virtualization) and device passthrough
- Virtual NICs and network bridges
- Virtual switches and network isolation

Network Protocols

01 QUIC (only available when we have ≥ 21 students)

- 02 HTTP/3
- 03 BitTorrent Protocol (only available when we have ≥ 25 students)

Modern RPC

- 01 Protobuf
- 02 Google RPC

Web-App Security

- 01 SOP (Same-Origin Policy), CORS (Cross-Origin Resource Sharing) (Foundations)
- 02 CORP / COOP / COEP (Cross-Origin Resource/Opener/Embedder Policies) (only available when we have ≥ 23 students)
- 03 CSP (Content Security Policy) and SRI (Subresource Integrity)

Introduction and concrete examples how they are used/specified and help prevent attacks.

Monitoring & Debugging Distributed Systems

- 01 Log Aggregation with a particular focus on correlation of log entries

Leader Election

- 01 Bully Algorithm and/or Ring Algorithm

Quorum Systems

- 01 Majority voting (i. e., quorum-distributed computing)

Consensus Algorithms and Fault Tolerance

- 01 Consensus Fundamentals & Problem Definition
  - What is consensus and why is it hard in distributed systems?
  - The FLP impossibility result (conceptual understanding)
  - Fault models: crash faults vs Byzantine faults
  - Safety vs liveness properties
  - Real-world motivation: replicated state machines, distributed databases
- 02 (Practical) Byzantine Fault Tolerance
  - When do we need BFT?
  - Modern developments
  - Real-world usage
- 03 Paxos Family
  - Basic Paxos algorithm (conceptual overview, roles: proposers, acceptors, learners)
  - Why Paxos is correct but complex
  - Multi-Paxos for practical systems
  - Real-world usage
- 04 Raft - Understandable Consensus
  - Motivation
  - Leader election, log replication, safety

- How Raft differs from Paxos (design philosophy)
- Real-world usage

Eventual Consistency

01 Eventual Consistency and Gossip Protocol

02 CRDTs (Conflict-free Replicated Data Types) (only available when we have ≥ 22 students)

Distributed File Systems

01 Ceph

02 HDFS (only available when we have ≥ 24 students)

Topic Assignment

The presentations need to be given in the order listed below.

Topic	Constraint	Student Name
01. Introduction to Virtualization & Use Cases		
02. Hypervisors - Architecture & Types		
03. Virtual Machines		
04. Containers & OS-level Virtualization		
05. Memory Virtualization		
06. Network & I/O Virtualization		
07. QUIC	≥ 21 students	
08. HTTP/3		
09. BitTorrent Protocol	≥ 25 students	
10. Protobuf		
11. Google RPC		
12. SOP, CORS (Foundations)		
13. CORP / COOP / COEP	≥ 23 students	
14. CSP and SRI		
15. Log Aggregation		
16. Leader Election: (a) Bully, (b) Ring Algorithm		
17. Quorum Systems (Majority Voting)		
18. Consensus Fundamentals & Problem Definition		
19. Byzantine Fault Tolerance (PBFT)		
20. Paxos Family		
21. Raft - Understandable Consensus		
22. Eventual Consistency and Gossip Protocol		
23. CRDTs	≥ 22 students	
24. Ceph		
25. HDFS	≥ 24 students	

Moderator Assignment

Topic Id	Moderated by Student with Topic Id
1	7
2	8
3	10
4	11
5	/
6	12
7	1
8	2
9	/
10	3
11	4
12	6
13	/

Topic Id	Moderated by Student with Topic Id
14	15
15	16a
16a	18
16b	22
17	19
18	16b
19	23
20	24
21	14
22	17
23	20
24	21
25	/

Presentations - Grading (max 25 Points)

- Presentation (max 22 Points)
- Checklist (max 1 Point)
- Moderation (max 2 Points)

  A Moderator has three main tasks:

  1. (Introduce the speaker) and *motivate the topic*.
  2. Keep track of time and *give time warnings to the presenter* (2 minutes left, time is up); abort the presentation if necessary (-1 minute).
  3. Lead a short Q&A session (2-3 questions) after the presentation; if there are no questions from the audience, *the moderator should have two to three questions*.

Programming Task (max 25 Points)

- Carried out in Groups of two students
- Grading will be done during an examination interview (15 Minutes per Group):
  1. Demonstration of your project (~5 Minutes)
  2. Answering specific questions related to your implementation / adapting your code to a new scenario
- Download here: 2026-04-06-Messaging-Project.zip

Lecture - Schedule

3. Mar 2026: Lecture

17. Mar 2026: Lecture Assignment of moderators to presentation topics (see above)

7. Apr 2026:
- Presentations:
  - 01. Introduction to Virtualization & Use Cases
  - 02. Hypervisors - Architecture & Types
  - 03. Virtual Machines
  - 04. Containers & OS-level Virtualization
  - 06. Network & I/O Virtualization
  - 07. QUIC
- Programming Task Explanation
- Lecture

21. Apr 2026 (5VL):
- Presentations (8 students)
  - 08. HTTP/3
  - 10. Protobuf
  - 11. Google RPC
  - 12. SOP, CORS (Foundations)
  - 14. CSP and SRI
  - 15. Log Aggregation
  - 16. Leader Election: (a) Bully, (b) Ring Algorithm
- Lecture

24. Apr 2026 (5VL):
- Presentations (8 students)
  - 17. Quorum Systems (Majority Voting)
  - 18. Consensus Fundamentals & Problem Definition
  - 19. Byzantine Fault Tolerance (PBFT)
  - 20. Paxos Family
  - 21. Raft - Understandable Consensus
  - 22. Eventual Consistency and Gossip Protocol
  - 23. CRDTs
  - 24. Ceph
- Lecture

26. April 2026, 07:00 (7:00 a.m.):

- Programming Task Submission Deadline

  Upload:

  - the JAR files as described in the task description
  - a ZIP file with the source code
  - a PDF file which explicitly states who worked on which part and which (optionally!) states whether you want to be graded as a group or individually w.r.t the submitted code. The oral exam is always graded individually.

27. April 2026:

- Oral Examination (Programming Task)

  Duration: 15 minutes

  During the oral examination, you will be asked to present, demonstrate, and explain your solution.

  The examination will focus on your solution to the programming task and the concepts directly related to it. It will not cover unrelated topics.

  You are required to bring your own notebook computer with a functioning IDE, all source code, and precompiled binaries. Your solution must be fully operational and ready to run on your device.

Exam Interviews

8:00	Philipp Weissgerber	Sebastian Schierholz
8:20	Lennart Kleemann	Ensar Sinanovic
8:40	Oskar Prophet	Faraz Kaeni
9:15	Lea Spannbauer	Lars Junge
9:35	Felix Wedel	Colin Kleinschmidt
9:55	Shmail Sheikh	Kai Bongartz
10:15	Alison Akongha	Siham Assem
10:40	Jannis Köllner	Jonas Andrae
11:00	Gabriel Klisanin	Hai Viet Vu
11:20	Tolgay Yildirim	Mert Bayrak
11:40	Hassan Idriss	Ruben Fitz